AI Tutor: System Design Diagrams

An Evidence-Based Adaptive Learning System with Deliberate Scaffolding Withdrawal

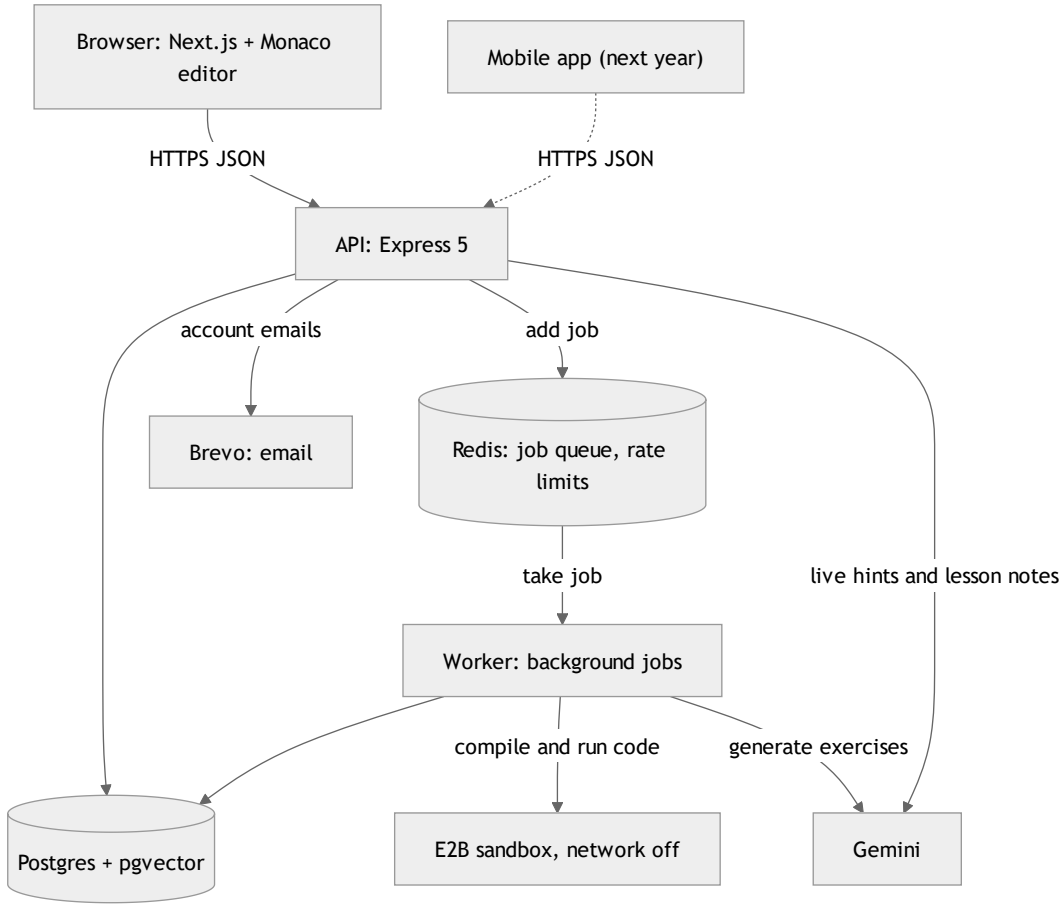
MCA-269 Minor Project, Jagan Institute of Management Studies, Rohini · Guide: Dr. Chetna Laroia

Ashwani Kumar Puri (04914004425) · Aman Dogra (01614004425) · Aditi Tanwar (01814004425)

Taken from the Phase 3 report (Chapter 3, System Design), 21 October 2026.

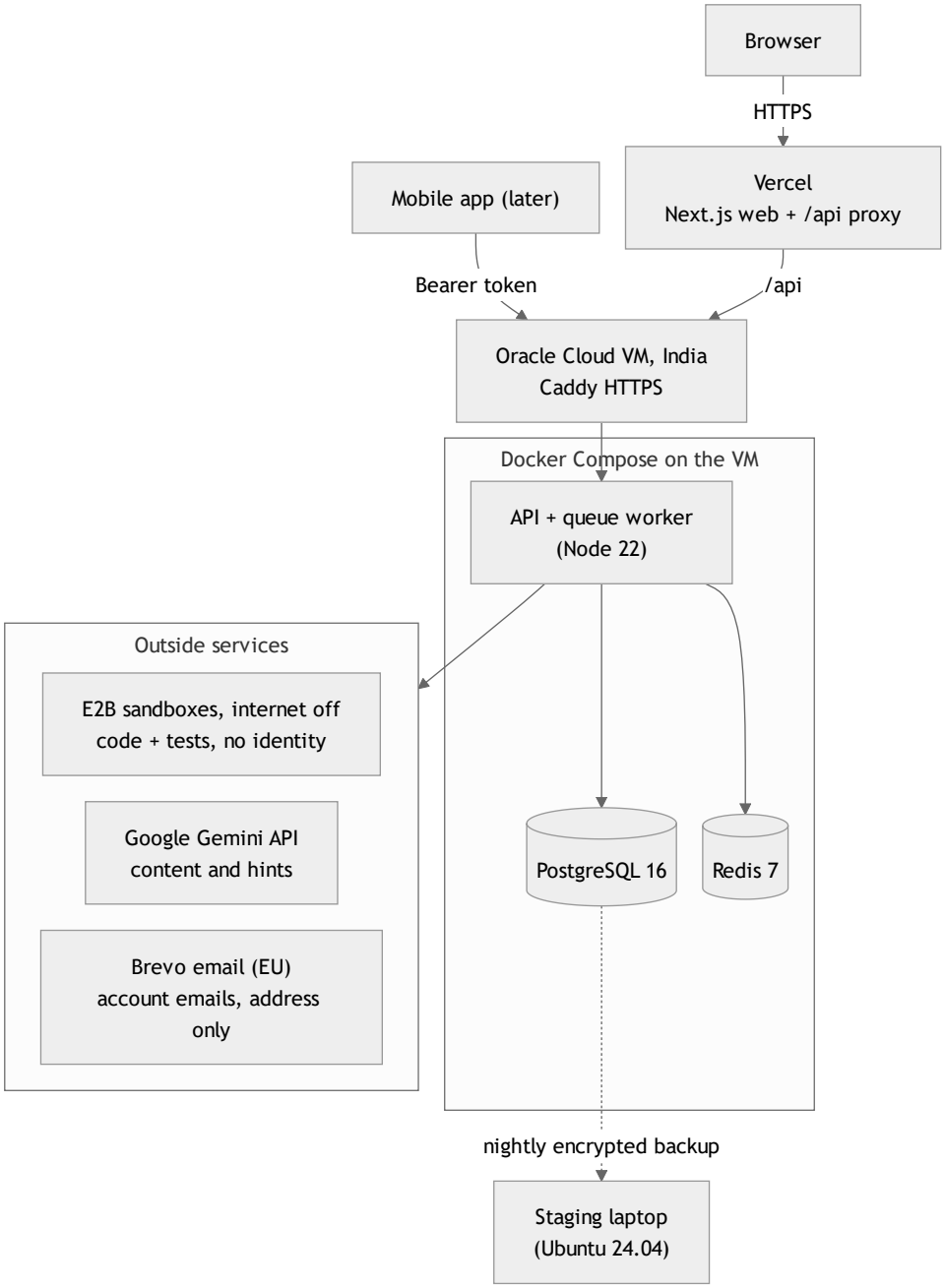
1. System Architecture

Figure 1: System architecture



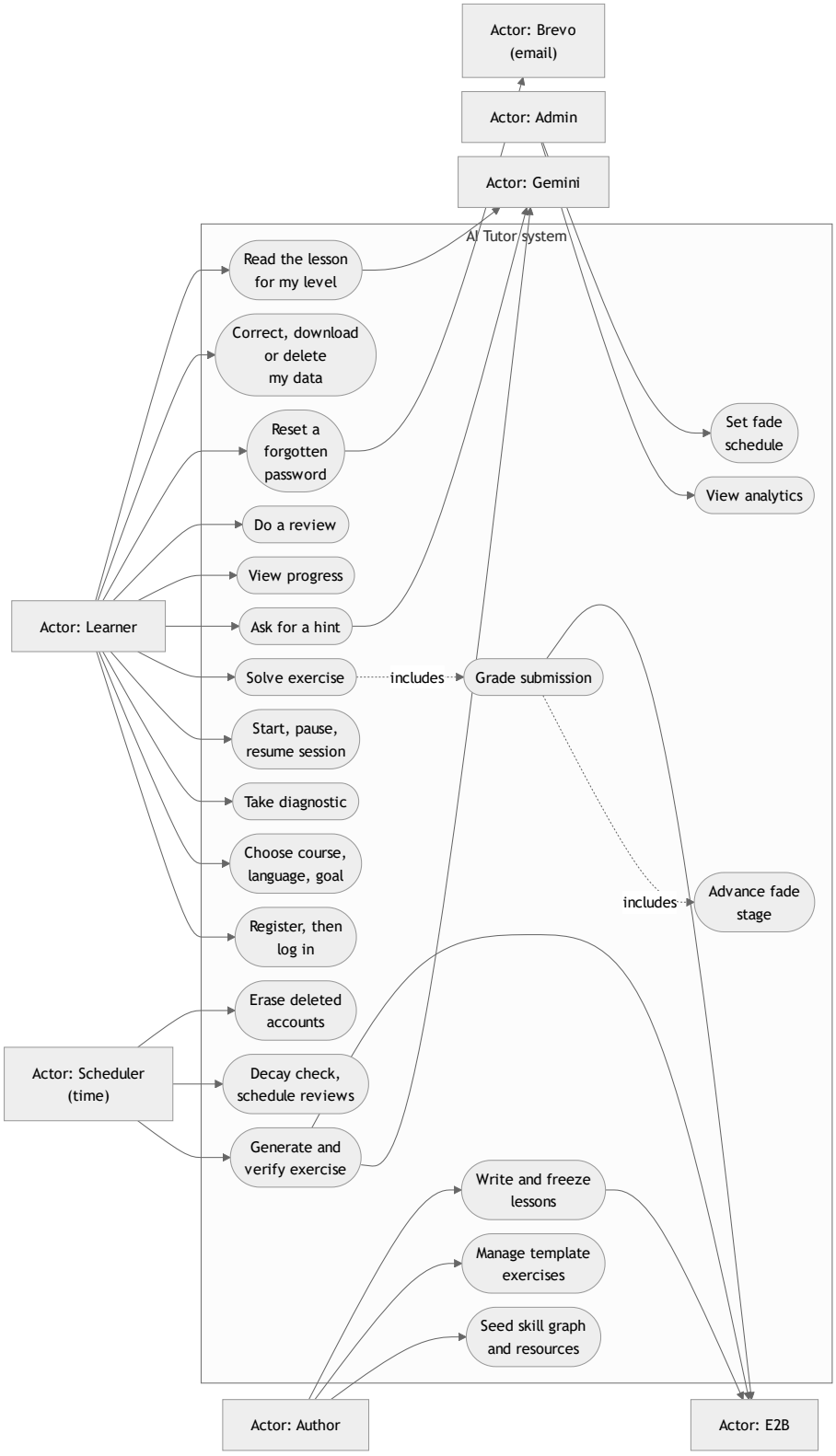
2. Deployment Diagram

Figure 12: Deployment



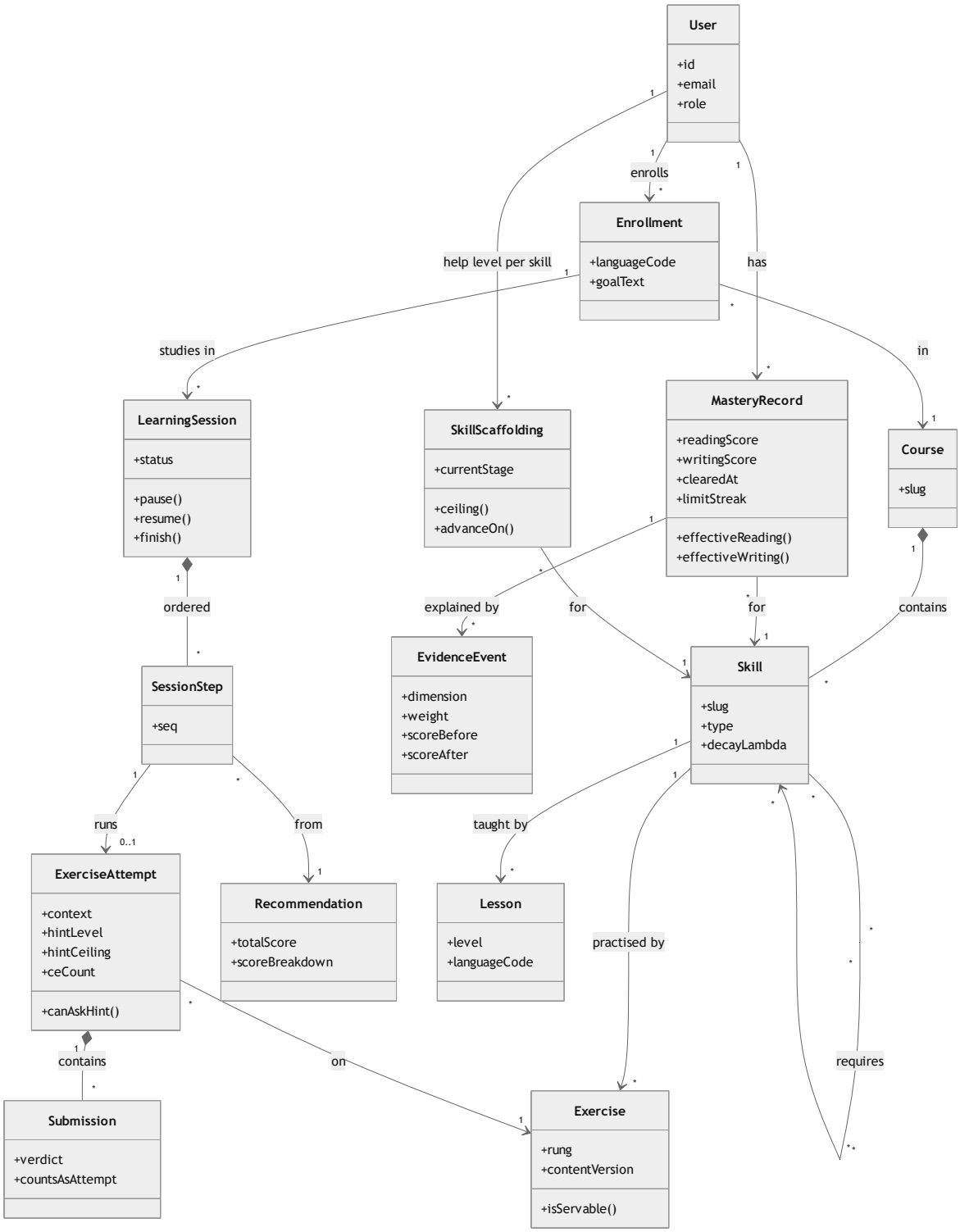
3. Use Case Diagram

Figure 2: Use case diagram



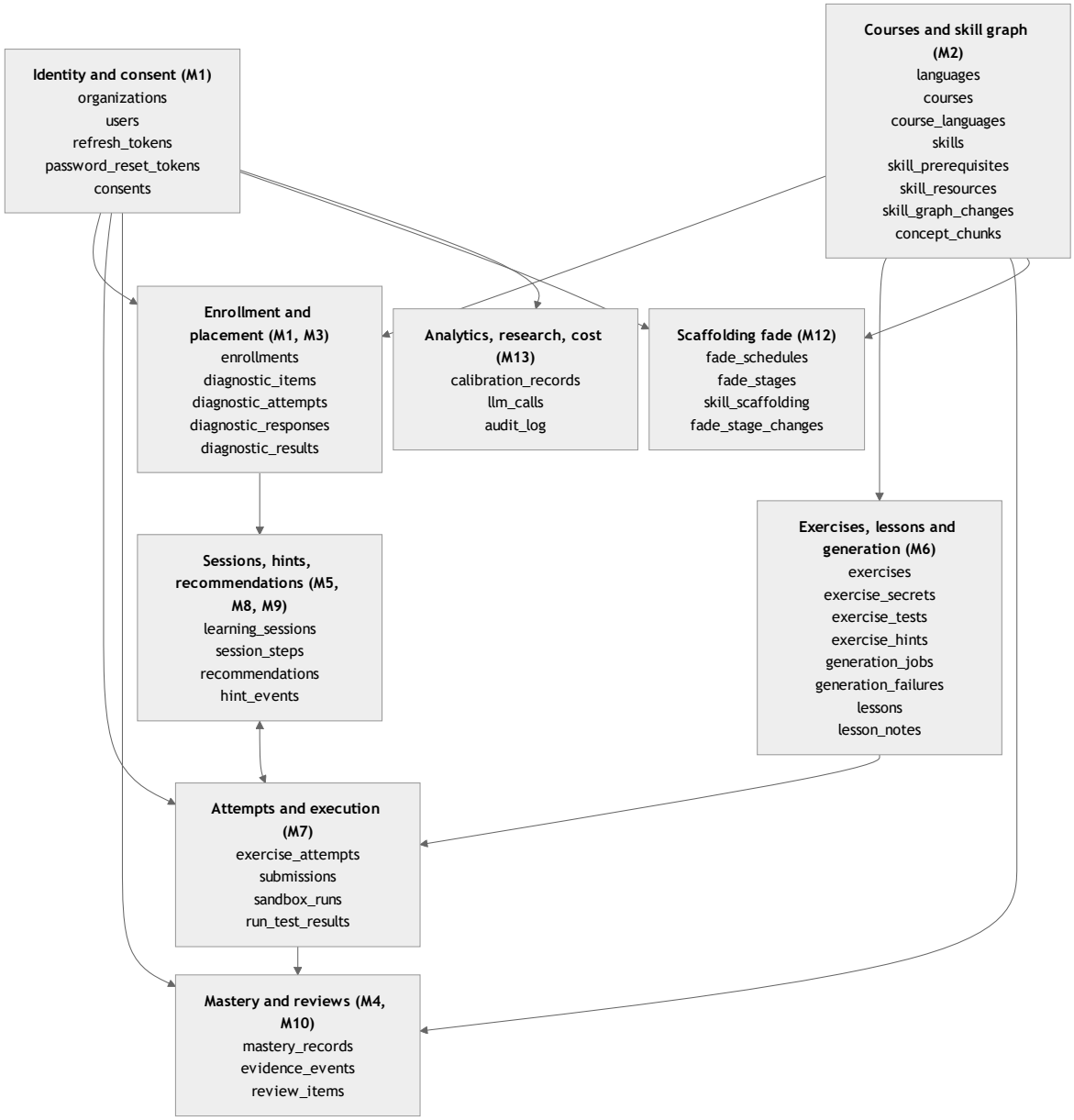
4. Class Diagram (Domain Model)

Figure 3: Domain model



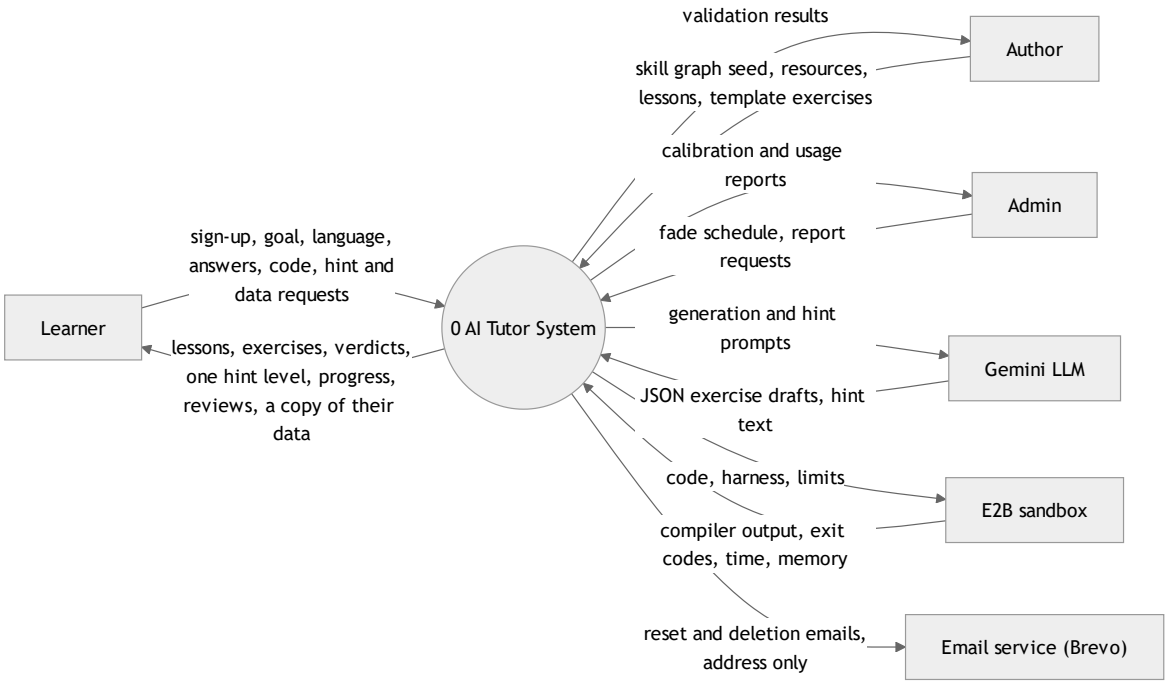
5. ER Diagram (Table Groups)

Figure 9a: Overview of the table groups.



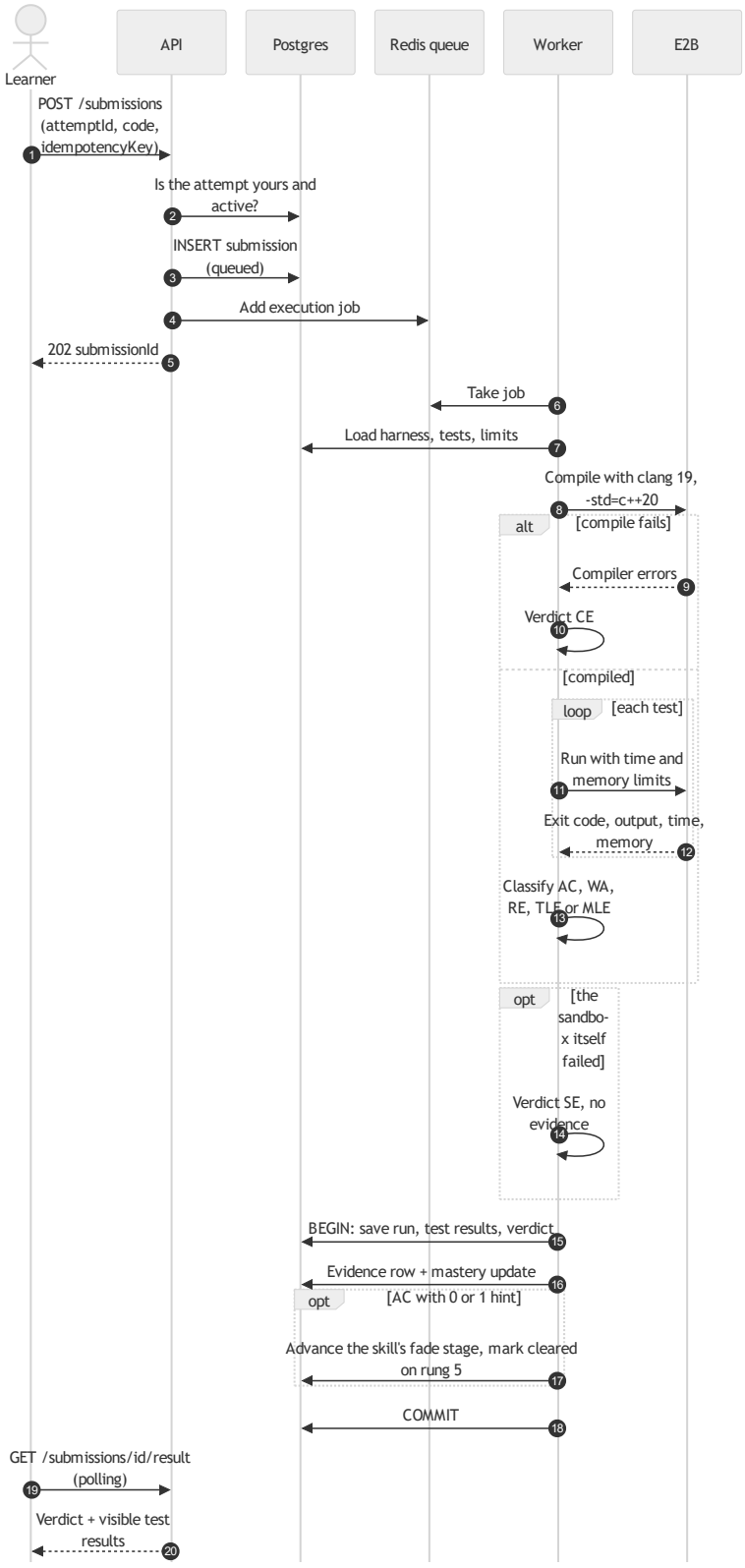
6. Data Flow Diagram, Level 0

Figure 10a: Level 0 (context)



7. Sequence Diagram: Submitting Code

Figure 6: Submit code.



8. Activity Diagram: Choosing the Next Step

Figure 8: Activity diagram, next-step selection, as built (recommender m5-v1, ADR-0011).

